QPCODE: 25900751

MAHATMAGANDHIUNIVERSITY,KOTTAYAM
MGU-UGP(HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

1st SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCBOT100, MG1DSCBBT100 - FASCINATING WORLD OF PLANT SCIENCES
(2024 ADMISSION ONWARDS)

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Studentsshouldattemptatleastonequestionfromeachcourseoutcometoenhancetheir overall Outcome attainability

Duration: 1 Hour 30 Minutes Max.Marks:50

Part A - Short Answer Questions Answer any 10 Questions - Each Question carries 1 mark			
1	What family does Quinine Tree belong to?		[K] / [CO1]
		Rubiaceae	
2	Identify the significance of superhydrophobic surfaces		[K] / [CO2]
		Superhydrophobic surfaces are significant because they repel water and have applications in self-cleaning, drag reduction, anti-icing, anti-corrosion, and oil-water separation	
3	How paper chromatography is typically carried out.		[K] / [CO3]
		Paper chromatography process separates mixtures based on the different affinities of compounds for the stationary phase (paper) and the mobile phase (solvent).	
4	What are the common reproduction methods in algae?		[K] / [CO2]
		Algae commonly reproduce through three main methods: vegetative, asexual, and sexual reproduction ***(any two methods – full mark)***	
5	Lichens are a symbiotic relationship between fungi and _____.		[K] / [CO2]
		Alga	
6	Which is the source of Taq DNA polymerase enyme?		[U] / [CO3]
		Taq DNA polymerase is sourced from a thermophilic bacterium called Thermus aquaticus.	
7	What is the main characteristic feature of gymnosperm seeds?		[K] / [CO2]
		The main characteristic of gymnosperm seeds is that they are "naked," meaning they are not enclosed within a protective fruit	
8	Illustrate different types of venations on leaves		[A] / [CO3]
		The two main types of leaf venation are parallel and reticulate	
9	Define GIS?		[K] / [CO3]
		A GIS, or Geographic Information System, is a computer system that captures, stores, analyzes, and displays geographically-referenced data.	
10	List the main parts of a compound microscope		[K] / [CO3]

		The main parts of a compound microscope are the optical parts (eyepiece, objective lenses) and the mechanical parts (base, arm, stage, and focus knobs).	
11	Who is considered as the "Father of Botany"?		[K] / [CO1]
		The ancient Greek scholar Theophrastus is known as the "Father of Botany"	
12	List any two applications of plant genetic engineering		[U] / [CO4, CO5]
		Two applications of plant genetic engineering are improving crop traits like pest resistance and nutritional value, and using plants as bioreactors to produce pharmaceuticals	
			[10x1 = 10]

Part B
Short Essay Type Questions
Answer any 6 Questions Each Question carries **5** marks

13	Explain the statement - all food comes from plants.		[A] / [CO1]
		Plants are the foundation of nearly every food chain. Plants produce their own food through photosynthesis, using sunlight, water, and carbon dioxide to create energy. Humans and animals get energy directly by eating plants, or indirectly by eating animals that have eaten plants. Plants are the ultimate food source - ❖ Directly: You eat food that comes directly from plants, such as fruits, vegetables, grains, nuts, and seeds. ❖ Indirectly: Animals that eat other animals (carnivores and omnivores) still ultimately rely on plants. In food chain - the energy captured in plants passes to the herbivores and then to carnivores. Similarly, milk and eggs come from animals that consume plants, making those products indirectly plant-based.	
14	Discuss the evolutionary trends leading to the dominance of angiosperms in plant diversity.		[A] / [CO2
		The dominance of angiosperms (flowering plants) is attributed to a suite of evolutionary innovations, particularly in their reproductive strategies, physiological efficiency, and ecological adaptability. These traits provided significant advantages over gymnosperms and ferns, leading to their rapid diversification and global spread. Key evolutionary trends and innovations include: ➢ Flowers and Animal Pollination ➢ Enclosed Ovules (Carpel) ➢ Fruits and Seed Dispersal ➢ Double Fertilization and Endosperm ➢ Efficient Vascular System ➢ Leaf Efficiency and Genome Size ➢ Growth Habit and Adaptability. ➢ Whole-Genome Duplications (WGDs): **(any five points with explanation in atleast one sentences – full mark)**	

15	How can a plant science scholar evaluate the extend of forests and vegetation patterns of a large land area?	[E] / [CO5]
	Evaluating the extent of forests and vegetation patterns over large land areas primarily relies on a combination of remote sensing technologies, Geographic Information Systems (GIS), and ground-based field surveys. These methods allow for consistent, cost-effective, and detailed monitoring over vast regions. **Key Indicators for Evaluation** The extent and patterns are typically assessed using metrics such as: ➢ Forest Area and Canopy Cover ➢ Species Composition and Biodiversity ➢ Biomass and Carbon Stocks ➢ Fragmentation and Connectivity ➢ Disturbance and Change Detection ***(liberal valuation)***	
16	Write a short note on plants living in extreme environments	[K] / [CO2]
	Plants in extreme environments, known as extremophiles, have developed diverse adaptations to survive harsh conditions like extreme temperatures, drought, and high salinity. These adaptations can be morphological (e.g., thick stems for water storage), physiological (e.g., producing antifreeze proteins), or anatomical (e.g., waxy coatings on leaves). **Adaptations for specific extreme environments** Desert environments: ➢ Water conservation:. ➢ Heat tolerance:. ➢ Drought tolerance ➢ Fire resistance:. Extreme cold: ➢ Ice prevention: ➢ Seasonal cycles: Plants. High salinity: ➢ Salt tolerance ➢ Salt excretion . ➢ Root protection Other adaptations ➢ Low nutrient availability: ➢ Deep roots:. ➢ Symbiotic relationships ***(OR - Plants survive in volcano, desert and arctic region-example (name) with at-least 2 adaptations- full mark).***	
17	Comment on the importance of botanical illustrations	[U] / [CO3]
	Botanical illustrations are important for scientific documentation, education, and conservation because they provide a detailed and accurate visual record of plants that is often more precise than a photograph. They are crucial for plant taxonomy, helping scientists identify and classify species by highlighting key diagnostic features, and serve as vital educational tools for students and the public. Illustrations document rare or endangered species before they disappear and promote a deeper appreciation for nature and its conservation.	

Scientific and educational importance

- Scientific documentation: Illustrations provide a clear and detailed record of plant morphology and anatomy
- Plant taxonomy: They have historically played a fundamental role in classifying plants and continue to help scientists compare species by emphasizing important features that may be missed in photographs.
- Education: used in textbooks, field guides, and other educational materials to teach plant biology and ecology to students and the general public.
- Documentation of new and endangered species: Artists can create precise illustrations of new species or record endangered plants before they are lost, ensuring that reference material survives for future generations.

18	Describe how smart devices can be used in agriculture to manage water stress?	[A] / [CO5]

Smart devices manage water stress in agriculture through sensor-based monitoring, data-driven analytics, and automated precision irrigation. IoT sensors track soil moisture, while AI and machine learning analyze real-time data from these sensors and weather forecasts to determine precise irrigation needs, leading to automated adjustments of systems like drip irrigation to deliver water only when and where it is needed.

How smart devises help to manage water stress

- ❖ Prevent water wastage: By using real-time data, systems deliver only the required amount of water, reducing waste from runoff, evaporation, and overwatering.
- ❖ Optimize crop health: Precision application of water and nutrients directly to the root zone improves plant health and can lead to higher crop yields.
- ❖ Increase efficiency: Automation reduces labor costs and the need for manual guesswork, allowing farmers to manage their fields more efficiently.
- ❖ Adapt to environmental changes: Integrating weather data allows the system to automatically adjust irrigation schedules based on rainfall, temperature, and humidity.

(OR - Explanation on the uses of IoT/AI in agriculture- Full mark)

19	Analyse the significance of fig leaves in lemur diets.	[An] / [CO2]

Fig leaves hold significant importance in the diets of many lemur species, particularly as a consistent and available food source that provides vital protein, especially during periods when preferred fruits are scarce.
Key Significance of Fig Leaves in Lemur Diets

- ❖ Nutritional Value: Fig leaves can provide essential nutrients, including protein,
- ❖ Year-Round Availability (Fallback Food): different fig species mature asynchronously, ensuring a more or less continuous supply of leaves and fruits throughout the year,
- ❖ High Consumption Rates: Some lemur species, both their fruits and leavesare consumed in large quantity
- ❖ Dietary Flexibility: The ability to switch between fruit-heavy and leaf-heavy diets depending on seasonal availability,
- ❖ Overcoming Plant Defenses: Plants often contain chemical deterrents like tannins and alkaloids. Lemurs that consume fig leaves show adaptations or tolerances to process these secondary compounds, though consumption might be limited past a certain threshold.
- ❖ Ecosystem Role: As major consumers of figs, lemurs play an important ecological role as seed dispersers

(OR - Explanation on zoopharmacognosy aspects- Full mark)

20	What are the differences between a TS and LS	[U] / [CO3]

TS (Transverse Section) and LS (Longitudinal Section) are two ways of cutting a specimen to view its internal structure. A TS cuts across the object's axis, showing a cross-section or "slice" of its width, while an LS cuts along the length, displaying the internal structure from end to end.

Feature	T.S	L.S.
Direction of cut	Perpendicular to the length, across the axis	Parallel to the length, along the axis
View of structure	Cross-sectional (like a slice)	Lengthwise (from top to bottom or side to side)
Example (plant stem)	A horizontal cut showing a circular cross-section with layers arranged in a circle	A vertical cut showing the layers from the top to the bottom

(5 × 6 = 30)

Part C
Essay Type Questions Answer any **1** Question
Question carries **10** marks

21 Give and account on the special adaptations in plants with reference to the types to heliotropisim and pseudocopulation in orchids. [U] / [CO2]

Plants have evolved sophisticated adaptations, such as heliotropism and pseudocopulation in orchids, to optimize survival and reproductive success in diverse environments.

Heliotropism in Sunflowers

Heliotropism in sunflowers is a fascinating example of how plants have evolved to adapt to their environment and can actively move to optimize their exposure to sunlight. Sunflowers, especially when they are young, exhibit a behavior called solar tracking, where their heads follow the movement of the sun across the sky during the day. By following the sun, young sunflowers maximize their energy intake, promoting better growth and ensuring reproductive success. Once mature, they continue to use their position to attract pollinators, completing the cycle of life.

East to West Movement:

- During the day, young sunflower plants face the east to catch the rising sun. As the sun moves across the sky, the sunflower's stem slowly bends to track the sun's path, turning from east to west. This movement is driven by the differential growth of cells on the stem. The side facing away from the sun grows faster, causing the plant to bend toward the light.
- At night, the sunflower "resets" by reorienting itself to face the east again.

Mechanism:

The movement is controlled by the plant's circadian rhythm, a natural internal clock that helps regulate its daily movements. Auxins, a type of plant hormone, play a crucial role in this process. They accumulate on the shaded side of the stem, promoting cell elongation, which causes the plant to bend toward the light.

Mature Sunflowers:

Once sunflowers reach maturity and their large heads develop, they stop moving. At this stage, the flowers generally remain facing east. It's believed that east-facing sunflowers warm up more quickly in the morning sun, making them more attractive to pollinators like bees, which prefer warmer flowers.

Benefits of Heliotropism in Sunflowers - Increased Photosynthesis: By tracking the sun, young sunflowers can maximize the amount of sunlight they absorb throughout the day. This increases their ability to photosynthesize, promoting faster growth and better energy production.

Pseudocopulation in Orchids:

Pseudocopulation in orchids is a remarkable example of evolutionary mimicry and deception. By mimicing the appearance, texture or scent of female insects, some orchid species deceive the male insects into attempting to

mate, thus ensuring pollination. This strategy ensures pollination without offering any nectar or other rewards to the pollinator. This strategy is highly energy-efficient and has led to intricate co-evolution between the orchids and their specific pollinators. This unique form of mimicry demonstrates the incredible adaptability of plants in their efforts to reproduce successfully. Example: Ophrys Orchids (Bee Orchids) and Drakaea Orchids (Hammer Orchids)

Advantages of Pseudocopulation for Orchids

- ❖ Energy Efficiency: Unlike many flowering plants that produce nectar to attract pollinators, orchids using pseudocopulation do not invest energy in producing food rewards. This makes pseudocopulation an efficient strategy, as the orchid can focus its resources on reproduction and growth rather than nectar production.

- ❖ Specialized Pollination: Pseudocopulation often results in species-specific pollination. The orchid is highly adapted to deceive one particular insect species, ensuring that pollen is reliably transferred between flowers of the same species. This reduces the risk of pollen being wasted on the wrong species of flower, increasing the likelihood of successful fertilization.

22	Describe the different types of microtomes and discuss their specific applications.	[U] / [CO3]

Microtomes slice samples into thin sections for microscopic examination, with different types designed for specific applications, such as rotary for routine histology, cryostats for frozen samples, sliding for large or hard blocks, and ultramicrotomes for ultra-thin sections required for electron microscopy. Other types include vibrating microtomes for fresh, soft tissue and saw microtomes for extremely hard materials like bone.

Types of microtomes and applications

Rotary microtome : The most common type, using a rotary motion to feed the sample towards a fixed knife.

- ➢ Application: Routine histology and pathology labs for sectioning paraffin-embedded tissues.

Cryostat microtome (Cryomicrotome) : A rotary microtome housed in a refrigerated chamber, typically at -20^0C - -30^0C

- ➢ Application: Rapid sectioning of frozen tissue samples, often used in surgical pathology and nerve biopsies.

Sliding microtome : The knife slides across a stationary block of tissue.

- ➢ Application: Sectioning large or particularly hard blocks of tissue, such as those embedded in celloidin, or for materials like wood, bone, and teeth.

Vibrating microtome (Vibratome) : Uses a vibrating blade to cut samples.

- ➢ Application: Sectioning soft, fresh, or unfixed samples without freezing or embedding, minimizing damage to cell morphology.

Ultramicrotome : A specialized instrument for producing ultra-thin sections.

- ➢ Application: Preparing sections as thin as 40–100 nm for transmission electron microscopy (TEM).

Saw microtome : Uses a saw for cutting.

- ➢ Application: Sectioning extremely hard and brittle materials like bone, teeth, and ceramics.

Laser microtome : A non-contact method that uses a laser to section samples.

- ➢ Application: Applications where a sample's cell morphology must remain unaltered, as it eliminates physical contact and minimizes sample distortion.

(Explanation of rotary, sledge and ultra microtome - full mark)

		(10 × 1 = 10)